

QuarkWave MIDI documentation

Audience: development testers using an assembled instrument, and developers extending it.
Status: source-reviewed, partial hardware check. These pages describe firmware behavior inspected on **2026-09-23**. A browser and USB smoke pass has been run on the two-board build; audio and end-to-end MIDI reception remain unverified.

Choose a guide

- [Quick start](#) — first note, first factory sound, and first saved variation on an assembled instrument.
- [Musician guide](#) — connect, play, shape sounds, use effects, and manage patches. Start here if you know synth controls but do not need firmware internals.
- [Patch book](#) — eight built-in sounds, what shapes each one, and five new patch recipes to try.
- [Sound-design lessons](#) — short listening exercises for oscillator, envelope, filter, movement, and effects.
- [Uno LED display](#) — read the top-row connection lights, voice bars, output meter, and waveform view.
- [External MIDI controllers](#) — play the Uno alone, send clock, and use its supported sound commands.
- [Technical guide](#) — board roles, signal flow, WebSocket and MIDI messages, patch files, and firmware behavior.
- [Diagram style](#) — visual conventions for maintaining the guide figures.
- [Connection guide](#) — photographs of the current USB-powered build, known board wiring, and proposed line and TDA1308 headphone connections.
- [Hardware test log](#) — repeatable checks and space for measured results.
- [Documentation gaps](#) — source discrepancies and tasks that need a product decision or hardware check before the guides can make stronger promises.

The **panel** images are simulated connected views of the current layout. They predate the 🧩 masthead mark; controls and positions are unchanged. They help locate controls but are not evidence of a tested two-board instrument. The [connection guide](#) contains owner-supplied photographs of the actual build; its line-output diagram is a proposed circuit, not an as-built photo.

Firmware snapshot

The two sketches and embedded browser page are the source of truth for these pages:

Component	Source	SHA-256
Uno R4 sound engine	QuarkWave_MIDI.ino	f2fc0d9a9740513331098414cb7556f49c951a5e8747a31c1a9fe2feb0c76ffc
Pico W web controller	QuarkWave_UI_MIDI.ino	7733641b8651cf2f9ba992490a02067783f2995ebb1b3a2d34e240dd6c84b794
Embedded browser panel	QuarkWave_UI.h	6975cecbde628688c93ce3d2146a6dbdc03b2443f6a4dde6f8d89fa7827cf8df

The connected controller identified over USB on 2026-09-23 is a **Pico W (RP2040, 2 MB flash)**. Its deployed build uses a 1 MB LittleFS region to match the existing patch storage. The Uno identified as **Arduino UNO R4 WiFi**. Both uploads verified; the live panel showed **Pico: Connected** and **Uno: Connected · Pico patch** with the existing “Test Pluck” user patch. An automated browser pass exercised 50 control paths and 41 pointer keys, plus all 12 computer keys at three velocities. The Pico reported changes and the Uno handshake was initially connected, but the Uno does not acknowledge each sound command. A Pico return-message parser fix resolved the reproduced explicit-sync failure in repeated two-board tests. Audio and per-control Uno sound response remain unverified. See the [hardware test log](#).

The older sketches under `../QuarkWave/` are outside this documentation baseline. The `secrets.h` files contain local Wi-Fi settings and are deliberately excluded.

Reading the status labels

Source-reviewed means the described path exists in this snapshot. **Hardware untested** on a specific procedure means that procedure still needs a two-board test. The handshake and browser paths with recorded results are identified in the [hardware test log](#); electrical levels and audio remain unmeasured. A procedure that cannot be established confidently from source is listed in [Documentation gaps](#) instead of being presented as a working feature.

When firmware changes

Recheck the three files above, update the protocol and control tables, walk through the musician tasks on the assembled unit, then change the snapshot date and hashes. Record unresolved changes in the gaps page.